

IAC-25,B3,9-GTS.2,2,x102920

The role of of Space Tourism and suborbital flight experiences in future astronaut selection criteria

Saira O. Williams^{a*}, Luísa Santos^b, A'laylah Morin^c, Aishwarya Kaushik^d, Dr. Rosanna Tilbrooke^e, Lielka Noelia Caballa-Huaman^f

^a *Space Generation Advisory Council, Schwarzenbergplatz 16, TOP 1 1010 Vienna, Austria*

^b *Space Generation Advisory Council, Schwarzenbergplatz 16, TOP 1 1010 Vienna, Austria*

^c *Space Generation Advisory Council, Schwarzenbergplatz 16, TOP 1 1010 Vienna, Austria*

^d *Space Generation Advisory Council, Schwarzenbergplatz 16, TOP 1 1010 Vienna, Austria*

^e *Space Generation Advisory Council, Schwarzenbergplatz 16, TOP 1 1010 Vienna, Austria*

^f *Space Generation Advisory Council, Schwarzenbergplatz 16, TOP 1 1010 Vienna, Austria*

^g *Universidad de Ingeniería y Tecnología (UTEC), Peru*

* Corresponding Author

Abstract

The landscape of human spaceflight is evolving rapidly with the rise of commercial space tourism and suborbital flight experiences. Traditionally, astronaut selection has been confined to highly trained professionals meeting rigorous physical, technical, and psychological criteria. However, the increasing frequency of private space missions has introduced new pathways to space, raising questions about how these experiences could influence future astronaut selection.

This paper focuses on examining the influence of space tourism and suborbital flight experiences on the evolving criteria for astronaut selection. Through an analysis of commercial spaceflight participation trends, mission performance data, and the backgrounds of space tourists and private astronauts, this study evaluates the potential role of suborbital and commercial orbital missions in astronaut preparation. The paper explores whether space tourism serves as a relevant training ground for future professional astronauts, assesses its impact on skills development, and considers how agencies and private spaceflight providers might integrate such experiences into future recruitment strategies. The findings aim to provide recommendations on how emerging spaceflight opportunities can contribute to astronaut training frameworks and inform evolving selection processes in an era of expanding human access to space.

Keywords: Space tourism; Suborbital flight; Astronaut selection; Commercial spaceflight; Human spaceflight; Training frameworks

Acronyms/Abbreviations

ISS – International Space Station
STEM – Science, Technology, Engineering, and Mathematics
ISRU – In-Situ Resource Utilization
CLG – Commercial Lunar Gateway
NASA – National Aeronautics and Space Administration
ESA – European Space Agency
ICAO – International Civil Aviation Organization
PAM – Private Astronaut Mission
PPH – Public-Private Harmonization
UBS – Union Bank of Switzerland
SGAC – Space Generation Advisory Council
UTEC – Universidad de Ingeniería y Tecnología

1. Introduction

The 21st century marks a turning point in human spaceflight, with commercial space tourism and suborbital flights reshaping access to space. Once limited to government-selected astronauts with military or scientific backgrounds, space travel is now accessible to private citizens, educators, researchers, and even artists. Since the first orbital tourist journey in 2001, the sector has expanded through initiatives by companies such as Blue Origin, Virgin Galactic, and SpaceX, enabling a wider demographic spectrum to experience microgravity, launch accelerations, and orbital operations [1–4].

These developments raise fundamental questions for astronaut selection: how should prior suborbital or commercial orbital experiences be valued in recruitment processes? What role can these experiences play in validating physiological and psychological resilience, team adaptability, or mission readiness? While critics argue that space tourism is primarily recreational, emerging evidence demonstrates that commercial participants undergo measurable stress adaptation and operational exposure, with potential parallels to agency-based astronaut training [5–11].

This paper explores the integration of commercial spaceflight into astronaut pathways, examining demographic diversification, training value, and standardisation challenges. By analysing trends in space tourism, mission performance, and institutional frameworks, the study evaluates whether suborbital and

orbital commercial missions should be recognised as preparatory steps for future astronauts. Ultimately, the paper argues that commercial experiences can contribute to a hybrid astronaut corps, bridging government and private sector expertise, provided that equity, certification, and global coordination are addressed.

2. Democratizing Access: The Rise of Space Tourism and Suborbital flights

Since 2001, dozens of private citizens have journeyed into space, primarily via suborbital tips or orbital missions to the International Space Station (ISS). This movement began with entrepreneur Dennis Tito, who paid approximately US \$20 million for a week-long trip aboard Soyuz TM-32 to the ISS in April 2001 (Wall, 2011). In the following years, a handful of wealthy individuals- including Mark Shuttleworth, Anousheh Ansari, Gregory Olsen, and Charles Simonyi (who flew twice)- followed in his footsteps, each paying tens of millions for orbital access.

The industry has since evolved beyond orbital tourism to include short-duration suborbital flights offered by companies like Blue Origin and Virgin Galactic. These offerings have increased annual participation, with the space travel market estimated to grow to more than US \$4 billion by 2030 (UBS, 2021).

One of the key enablers of this trend has been the technological and economic advances in reusable launch systems. Some notable players in this field include Blue Origin's *New Shepard* and Virgin Galactic's *VSS Unity*. Though still expensive, the reduction in per-flight costs and rise in operational frequency have enabled new classes of space travellers, many of whom would not have qualified under traditional astronaut selection criteria. As of August 2025, *New Shepard* has flown 75 individuals with 14 human missions in total (Blue Origin, 2025a), while Virgin Galactic inaugurated its first commercial suborbital flight in June 2023 (Virgin Galactic, 2023a).

Crucially, commercial spaceflight has driven greater diversity among participants, in contrast to the historically homogeneous profile of government-selected astronauts, who were predominantly white, male, military-trained, and from

STEM fields (Gres et al. 2023). For example, Virgin Galactic's Galactic 02 mission in August 2023 marked several milestones: the first female astronauts from the Caribbean, the first mother-daughter duo in space, and the largest number of women on a single commercial mission (Virgin Galactic, 2023b). Blue Origin's NS-31 mission in 2025 further advanced representation by flying the first all-female crew since Valentina Tereshkova's solo spaceflight in 1963 (Blue Origin, 2025b).

Beyond gender, missions are showcasing age and nationality diversity. For example, Anastatia Mayers, who flew on Virgin Galactic's Galactic 02 mission in August 2023 at just 18 years old, became the youngest person to travel to space (Boex, 2024), while Ed Dwight, who flew on Blue Origin's NS-25 mission in May 2024 at 90 years old, became the oldest (Yang, 2024). Commercial spaceflight has also enabled national firsts: notable examples include Sara Sabry on Blue Origin's NS-22 in 2022, who became the first Egyptian in space; Namira Salim on Virgin Galactic's Galactic 04 in 2023, who became the first Pakistani; Chun Wang on Fram2 in 2025, who became the first Maltese; and Owolabi Salis on Blue Origin's NS-25 in 2024, who became the first Nigerian [10]. These milestones highlight how space tourism is expanding global representation and access.

Space tourism has also diversified not only who goes to space, but why. SpaceX's *Inspiration4* mission in September 2021, the first all-civilian orbital flight, raised over US \$200 million for St. Jude Children's Research Hospital while conducting microgravity experiments with a crew composed of a geoscientist, physician assistant and cancer survivor, and aerospace data engineer (Kluger, 2021). Similarly, the planned *dearMoon* mission aboard SpaceX's Starship, although cancelled due to developmental delays, was designed to carry artists, filmmakers, and musicians on a circumlunar mission to the Moon for free, emphasizing creative and humanistic objectives over technical mandates (Howell, 2022). These missions illustrate how commercial spaceflight is reshaping not just participation but also purpose, broadening the narratives of exploration.

Taken together, these examples underscore how space tourism and suborbital flight are enabling a broader

demographic spectrum to access space- across gender, nationality, age, profession, and purpose. This trend not only enriches the cultural and social significance of space exploration but also raises critical questions for astronaut selection: how might non-traditional experiences and diverse backgrounds contribute to the success of future long-duration missions and crew dynamics?

3. Evaluating Readiness: What Do Tourists Really Learn?

The expansion of commercial space tourism and suborbital flights raises a critical question: to what extent do participants acquire competencies relevant to astronaut readiness? While these missions are often framed as recreational or symbolic, they expose participants to many of the physiological and psychological conditions as in professional spaceflights, although for shorter durations.

- **Physiological adaptation:** Suborbital flights by Blue Origin and Virgin Galactic subject participants to high G-loads during launch and re-entry, as well as short periods of microgravity lasting between 3 to 6 minutes. These experiences offer measurable opportunities to assess motion sickness susceptibility, vestibular adaptation (related to neuroplasticity), and cardiovascular responses, which are comparable to data collected from astronaut centrifuge training and parabolic flights [35]. Such real-world exposure contributes to assessing an individual's resilience under extreme conditions, a factor that cannot be fully replicated in simulations on Earth.
- **Psychological and cognitive readiness:** Beyond physical stressors, commercial spaceflight introduces participants to the psychological intensity of high-leverage missions. Studies in space psychology emphasize that composure, decision-making under stress, and emotional regulation are critical indicators of astronaut performance [16]. Findings on suborbital participants suggest that even limited-duration missions can act as a test framework for psychological readiness, offering parallels with early 'shuttle' era short-duration missions [36]. Furthermore, the widely reported "overview

effect," a cognitive shift described by many astronauts, has also been documented among space tourists, with the potential to foster long-term commitment to space activities and public advocacy [37].

- **Training value:** Despite these benefits, the scope of learning in tourism flights remains constrained. Suborbital missions lack the sustained operational demands of orbital missions, such as spacecraft maintenance, prolonged team coordination, and adaptation to extended periods of microgravity [36]. Thus, while suborbital experiences help evaluate basic physiological and psychological indicators, they represent a complementary preparation rather than being equivalent to formal astronaut training.
- **Symbolism and utility:** Importantly, the incursion of diverse participants into space tourism missions demonstrates that non-professionals can endure and even thrive under authentic spaceflight conditions. This suggests a new paradigm where space tourism serves also as a gateway to democratize astronautics, while providing agencies with valuable datasets on civilian adaptation to space. To leverage this potential, a structured framework should be established for standardized physiological and psychological monitoring during commercial missions, ensuring that collected data can directly inform and enhance future astronaut selection criteria. [38].

Although space tourists do not acquire the full operational expertise of professional astronauts, their flights provide valuable insights into human adaptation to space, help validate markers of readiness, and open new and more inclusive avenues toward astronautics.

4. Bridging the Gap: Integrating Commercial Experience into Astronaut Pathways

4.1 Reframing Commercial Spaceflight as a Strategic Asset

The emergence of commercial spaceflight is fundamentally reshaping the human spaceflight ecosystem, moving beyond tourism to create a new paradigm for astronaut readiness. As Crowther

emphasizes, this sector challenges established regulatory frameworks yet simultaneously generates new opportunities for skill development and operational validation [13]. Within Edgell's framework of sociotechnical transitions, from state-led programs (Space 1.0) to hybrid public-private partnerships (Space 3.0), commercial missions now function as strategic assets in astronaut preparation. Suborbital flights by Virgin Galactic and Blue Origin, along with orbital missions by SpaceX and Axiom Space, expose participants to G-forces, microgravity, and mission protocols, effectively producing a pool of *pre-screened candidates* with validated physiological and psychological resilience [14].

4.2 The Value Proposition: Physiological and Psychological Screening

Traditionally, astronaut candidates were selected for technical expertise, with spaceflight proficiency developed post-selection. Commercial spaceflight introduces a paradigm shift: pre-selection exposure to authentic space conditions. Suborbital flights provide an integrated test of launch accelerations, microgravity adaptation, and re-entry stresses, offering measurable validation of resilience [15]. Psychological benefits are equally significant, as participants demonstrate composure under the anticipation and stress of flight, traits strongly aligned with professional astronaut psychological assessments [16]. Orbital missions go further, delivering condensed yet comprehensive training on spacecraft systems, emergency procedures, and orbital mechanics [17], thereby reducing the baseline training burden for space agencies.

4.3 Differentiated Training Value: Suborbital vs. Orbital Missions

The preparatory value of commercial experiences varies with mission profile:

- **Suborbital flights** (Blue Origin, Virgin Galactic) provide high-stress launch and re-entry adaptation and a short period of microgravity. Their primary role lies in validating physiological thresholds and rapid adaptation [18].
- **Orbital missions** (SpaceX's *Inspiration4*, Axiom's Ax-2) offer extended experience in

microgravity operations, habitability management, and team coordination. Inspiration4's biomedical program, for instance, produced unprecedented datasets on rapid civilian adaptation to space, now informing NASA's human research programs [20]. Such missions cultivate operational competence and scientific engagement that parallel early stages of ISS expeditions [19].

4.4 Institutional Integration: From Participation to Pathway

Agencies face the challenge of systematically integrating this emerging talent pool. Several models are under discussion:

1. **Preferred Experience Model** – Treating completion of a certified orbital mission as a “preferred” qualification, analogous to flight hours for pilots [21].
2. **Partnership and Fast-Track Model** – Frameworks such as NASA's Private Astronaut Mission (PAM) to the ISS enable identification of high-performing participants for accelerated agency consideration [22].
3. **Modified Criteria Model** – Selection committees could incorporate structured debriefs of commercial mission experience, mirroring the use of analogous fieldwork in astronaut selection [23][30][38].

Within Edgell's Public-Private Harmonization (PPH) model, fixed-price contracts empower agencies to leverage commercial datasets for candidate evaluation, potentially reducing training redundancies and strengthening evidence-based screening [14].

4.5 Challenges of Equity and Regulation

The primary challenge remains equity. Commercial missions are prohibitively expensive, raising concerns about exacerbating socio-economic divides [18-Sample, 2021). Without policy safeguards, astronaut pathways risk privileging the wealthy. Moreover, training rigor varies across providers; agencies must therefore certify programs to ensure alignment with core astronaut competencies. Standardized accreditation,

similar to aviation licensing, will be essential to guarantee quality and interoperability [13].

4.6 Enabling Interplanetary Ambitions

Beyond individual preparation, commercial spaceflight accelerates system-level readiness for deep-space exploration. Edgell's Interplanetary Ambitions period emphasizes leveraging commercial innovations such as reusable launch vehicles and in-situ resource utilization (ISRU) [14]. NASA's Artemis program already depends on commercially developed lunar landers, while Commercial Lunar Gateways (CLGs) are envisioned as modular infrastructures for Mars preparation. In this ecosystem, commercial astronauts will increasingly form a hybrid workforce, bridging tourism, research, and exploration.

4.7 Toward a Hybrid Astronaut Corps

Commercial spaceflight experiences constitute a transformative variable in astronaut selection and training. They provide empirical screening, foundational competencies, and system-level innovations that complement traditional agency programs. To harness these benefits responsibly, agencies must adopt standardized accreditation, equitable access mechanisms, and collaborative regulatory frameworks. By doing so, commercial spaceflight can evolve from a recreational novelty into a cornerstone of astronaut development. The result will be the emergence of a **hybrid astronaut corps**, a synthesis of governmental and commercial expertise, positioned to sustain humanity's interplanetary trajectory.

5. Toward Standardization: Building a Global Framework

Suborbital space tourism is a catalyst for space industry revenue that could reach \$1.1 billion by 2032 [25]. This increased growth of human spaceflight access through commercial space tourism and suborbital missions underscores the urgent requirement for a harmonised global framework for astronaut selection and training. While traditional government programs (e.g., NASA, ESA, Roscosmos) have historically relied on highly selective standards rooted in stringent medical, psychological, and technical requirements, these frameworks were designed for professional astronauts

with long-term mission objectives (Deiss). By contrast, the emergence of commercial spacefarers—often with non-technical or varied professional backgrounds—introduces unprecedented diversity in astronaut profiles as mentioned above. This diversification challenges existing models to reconcile safety and operational readiness with the inclusive ethos of commercial spaceflight. A globally standardised framework would therefore serve to align governmental and private sector practices, ensuring mission success while enabling wider participation in human space exploration.

5.1 *The existing models and gaps*

At present, astronaut selection standards remain agency- or nation-specific, reflecting legacy criteria and institutional priorities. For example, NASA continues to emphasise advanced educational attainment, professional expertise (e.g., in aviation or STEM fields), and cumulative pilot flight hours, complemented by extensive medical and psychological evaluations. Similarly, ESA adheres to internationally recognised medical screening procedures for long-duration missions, while Roscosmos integrates its own physiological and operational benchmarks [28]. Commercial operators such as Blue Origin, SpaceX, and Virgin Galactic have adapted portions of NASA's safety and medical guidelines, yet no unified accreditation system exists across providers. This creates disparities in participant preparation, risk assessment, and operational training. For instance, while one provider may mandate centrifuge testing or hypoxia training, another may only require a medical self-certification [33]. These inconsistencies highlight the lack of cross-recognition between commercial flight experiences and agency-validated astronaut pathways, leaving a gap in standardised risk management and safety assurance.

5.2 *The components of a global framework*

A comprehensive **global standardisation framework** would require multi-dimensional integration of medical, technical, and ethical domains. Its core components should include:

1. **Medical and psychological criteria** validated against the unique stressors of spaceflight, including G-force tolerance, radiation exposure thresholds, and resilience to isolation and confinement [34].
2. **Training requirements** tailored to mission-specific roles, distinguishing between tourist participants, research astronauts, and mission specialists, while mandating baseline competencies in space physiology, emergency procedures, and spacecraft systems awareness. [23][30][38].
3. **Certification and accreditation mechanisms** to ensure commercial spaceflight programs are benchmarked to internationally recognised standards, thereby enabling seamless integration of commercial experience into professional astronaut corps pathways.
4. **Ethical and inclusivity guidelines** to advance diversity across gender, nationality, age, and ability, while ensuring adherence to non-negotiable safety and mission readiness standards [29].
5. **Safety and interoperability mandates** encompassing spacecraft design, operational protocols, and rescue capabilities, ensuring compatibility for joint or multinational missions [32].

5.3 *Implementation of collaboration amongst Institutions*

The establishment of a **global astronaut selection framework** would necessitate extensive **multilateral collaboration**. Agencies, commercial operators, regulatory authorities, and international organisations must collectively participate in governance. Possible avenues include:

1. **Formation of an international body** analogous to the International Civil Aviation Organization (ICAO), tasked with developing global astronaut certification standards, training protocols, and safety oversight mechanisms.

2. **Data-sharing platforms and research collaborations** to consolidate findings on the physiological and psychological effects of spaceflight, enabling refinement of selection benchmarks and risk thresholds.
3. **Harmonisation of national regulations**, such that astronaut qualifications and certifications are mutually recognised across borders, supporting interoperability and joint mission participation.

Such institutional coordination would establish a foundation for long-term global standardisation, while simultaneously enabling flexibility for innovation in training and technology.

5.4 Major challenges with the practice of a Global framework

Despite the clear rationale, several barriers to global standardisation persist. Chief among them are the sovereignty concerns of national space programs, particularly regarding control over astronaut selection and mission readiness. The financial costs of establishing and maintaining standardised commercial astronaut training also pose challenges, potentially limiting accessibility to wealthier participants or nations. Furthermore, ensuring equitable access for underrepresented groups remains a pressing concern in the face of commercialisation [31].

Nevertheless, the emergence of commercial spaceflight presents unique opportunities. It offers a scalable platform for preliminary training, screening, and skill development, thereby easing the transition into professional astronaut roles. By expanding the pool of individuals exposed to spaceflight stressors, commercial missions can accelerate the readiness of future crews for long-duration deep space exploration (e.g., lunar or Martian missions). This dual role—as both risk factor and enabler—positions commercial spaceflight as a catalyst for evolving astronaut selection paradigms.

6. Conclusions

Commercial spaceflight has transitioned from novelty to a strategic element in human spaceflight development. Suborbital and orbital missions provide unique opportunities to validate resilience, expose

participants to authentic space conditions, and generate data with direct applicability to astronaut training. The evidence reviewed indicates that these experiences can reduce training burdens, diversify astronaut profiles, and accelerate readiness for long-duration missions.

However, challenges remain. Without global frameworks for accreditation and safety standards, disparities in preparation and access risk undermining progress. Equitable participation and regulatory oversight are therefore essential to transform commercial spaceflight into a sustainable component of astronaut pathways.

In conclusion, the rise of space tourism and suborbital flight experiences represents more than an expansion of access. It is a catalyst for redefining astronaut selection, creating hybrid career paths, and shaping the future workforce of interplanetary exploration. Integrating these experiences responsibly into astronaut frameworks will help ensure that humanity's expansion into space is both inclusive and operationally effective.

Acknowledgements

The authors, as part of the DIVINAS research team under the Space Generation Advisory Council (SGAC), gratefully acknowledge the dedication and collaborative spirit of all team members who contributed to this work. The insights, discussions, and shared commitment to advancing inclusive and innovative approaches to human spaceflight were essential in shaping this study. We also thank the wider SGAC community for fostering a platform where emerging researchers can engage with critical questions at the intersection of space tourism, astronaut selection, and future exploration.

References

- [1] M. Wall, "First Space Tourist: How a U.S. Millionaire Bought a Ticket to Orbit," *Space.com*, Apr. 27, 2011.
- [2] UBS, "Future of Space Tourism: Lifting off? Or has COVID-19 stunted adoption?," Jul. 20, 2021, <https://www.ubs.com/global/en/investment-bank/insight-s-and-data/2021/space-tourism.html>, (accessed 09.09.25).

- [3] Blue Origin, “Blue Origin’s New Shepard Completes 34th Mission to Space,” Aug. 2025, <https://www.blueorigin.com/news/new-shepard-ns-34-mission>, (accessed 09.09.25).
- [4] Virgin Galactic, “Virgin Galactic Completes Inaugural Commercial Spaceflight,” Jun. 2023, <https://www.virgingalactic.com/news/virgin-galactic-completes-inaugural-commercial-spaceflight>, (accessed 09.09.25).
- [5] T. Gres, et al., “Study of the astronaut’s profile evolution since 1961: What makes a good astronaut since then and how did society impact it?,” Presented at the International Astronautical Congress, Baku, 2023.
- [6] Virgin Galactic, “Virgin Galactic Completes First Private Astronaut Spaceflight ‘Galactic 02’,” Aug. 2023, <https://www.virgingalactic.com/news/virgin-galactic-completes-first-private-astronaut-spaceflight-galactic-02>, (accessed 09.09.25).
- [7] Blue Origin, “Blue Origin’s New Shepard Completes 31st Mission to Space,” Apr. 2025, <https://www.blueorigin.com/news/new-shepard-ns-31-mission>, (accessed 09.09.25).
- [8] C. Boex, “Who is the youngest person in space? Depends who you ask,” *LiveNOW from FOX*, Aug. 30, 2024.
- [9] M. Yang, “First Black astronaut candidate, now 90, reaches space in Blue Origin flight,” *The Guardian*, May 19, 2024.
- [10] “List of space travelers by nationality,” *Wikipedia*, https://en.wikipedia.org/wiki/List_of_space_travelers_by_nationality, (accessed 09.09.25).
- [11] J. Kluger, “Inspiration4 Makes Space History With First-Ever All-Civilian Orbital Launch,” *TIME*, Sept. 15, 2021.
- [12] E. Howell, “Meet the dearMoon crew of artists, athletes, and a billionaire riding SpaceX’s starship to the moon,” *Space.com*, Dec. 26, 2022.
- [13] R. Crowther, “The regulatory challenges of ensuring commercial human spaceflight safety,” *Space Policy*, vol. 27, no. 1, pp. 74–76, 2011.
- [14] R. A. Edgell, “Sociotechnical pathways: From satellites and stations to envisioning commercial lunar gateways and beyond,” *AIAA SciTech Forum*, vol. 2025, no. 1, pp. 1–26, 2023.
- [15] Blue Origin and NASA JSC Flight Opportunities, *Suborbital Spaceflight Astronaut Certification*, Industry White Paper, 2020.
- [16] N. Kanas and D. Manzey, *Space Psychology and Psychiatry*, 2nd ed., Dordrecht: Springer, 2008.
- [17] A. Harrington, et al., “The Inspiration4 crew training overview,” SpaceX & Inspiration4 Press Kit, 2022.
- [18] K. Chang, “Branson’s flight opens a new era of space tourism. But for whom?,” *The New York Times*, Jul. 20, 2021.
- [19] Axiom Space, *Ax-2 Mission Overview*, Axiom Space, 2023.
- [20] J. Scott, et al., “Molecular and physiological changes in the Inspiration4 crew during spaceflight,” *Nature*, Pre-print, 2023.
- [21] ESAsci, “The European Astronaut Centre’s Space Medicine Office – Preparing for the Future,” European Space Agency, 2021.
- [22] NASA, *NASA’s Commercial Low-Earth Orbit Development Program*, NASA Fact Sheet, 2022.
- [23] L. B. Landon, K. J. Slack, and J. D. Barrett, “Teamwork and collaboration in long-duration space missions: Going to extremes,” *American Psychologist*, vol. 73, no. 4, pp. 563–575, 2017.
- [24] I. Sample, “Space tourism: Rocket billionaires launching a new cosmic era,” *The Guardian*, Jul. 11, 2021.
- [25] B. T. Musselman, S. R. Winter, S. Rice, J. R. Keebler, and K. J. Ruskin, “Point-to-point suborbital space tourism motivation and willingness to fly,” *Annals*

of *Tourism Research Empirical Insights*, vol. 5, 100119, 2025.

[26] H. Deiss, “Astronaut Requirements,” NASA, 2020, <https://www.nasa.gov/humans-in-space/astro-naut-requirements/>.

[27] SpaceRef Editor, “Principles Regarding Processes and Criteria for Selection, Assignment, Training and Certification of ISS Crewmembers,” *SpaceRef*, 2002, <https://spaceref.com/status-report/principles-regarding-processes-and-criteria-for-selection-assignment-training-and-certification-of-iss-expedition-and-visiting-crewmembers/>.

[28] “ESA - Astronaut selection 2021–22 FAQs,” European Space Agency, n.d., https://www.esa.int/About_Us/Careers_at_ESA/ESA_Astronaut_Selection/Astronaut_selection_2021-22_FAQs, (accessed 13.09.25).

[29] S. S. Kumar, “Space tourism: An analysis of its legal aspects,” *International Journal of Law*, vol. 7, no. 1, pp. 40–47, Jan. 2021.

[30] L. B. Landon, C. Rockholt, K. J. Slack, and Y. Pecena, “Selecting astronauts for long-duration exploration missions: Considerations for team performance and functioning,” *Acta Astronautica*, vol. 5, pp. 33–56, Mar. 2017.

[31] S. Singh, “Beyond Earth: Legal Challenges and Regulatory Frameworks in the Era of Space Tourism,” *International Journal for Legal Research and Analysis*, vol. 2, no. 7, Jan. 2025.

[32] L. Tedeman and K. Wright, “International spaceflight crew rescue standards,” *Acta Astronautica*, vol. 30, pp. 85–92, Jul. 1993.

[33] D. Webber, “A Training Framework for Private Space Travel Operations,” *International Astronautical Congress*, Naples, Italy, 63rd IAC, 2012.

[34] H. Yasuda and L. Sihver, “Broadening the selection criteria for Astronauts undertaking long-term space travel,” *Frontiers in Nuclear Medicine*, vol. 2, 997718, Nov. 2022.

[35] Jones, C. W., Overbey, E. G., Lacombe, J., Ecker, A. J., Meydan, C., Ryon, K., Tierney, B., Damle, N., MacKay, M., Afshin, E. E., Foox, J., Park, J., Nelson, T. M., Suhail Mohamad, M., Byhaqui, S. G. A., Aslam, B., Tali, U. A., Nisa, L., Menon, P. V., ... Mason, C. E. (2024). *Molecular and physiological changes in the SpaceX Inspiration4 civilian crew*. *Nature*, 632(8027), 1155–1164.
<https://doi.org/10.1038/s41586-024-07648-x>

[36] Chang, Y.-W., & Chern, J.-S. (2016). *Ups and downs of space tourism development in 60 years from moon register to spaceshiptwo CRASH*. *Acta Astronautica*, 127, 533–541.
<https://doi.org/10.1016/j.actaastro.2016.06.008>

[37] White, F. (2014). *The overview effect: Space exploration and human evolution* (3rd ed.). American Institute of Aeronautics and Astronautics.

[38] Landon, L. B., Slack, K. J., & Barrett, J. D. (2018). *Teamwork and collaboration in long-duration space missions: Going to extremes*. *The American Psychologist*, 73(4), 563–575.
<https://doi.org/10.1037/amp0000260>